



**Department of Physical Sciences**  
**Faculty of Applied Sciences**  
**Rajarata University of Sri Lanka, Mihintale**  
**B.Sc. in Applied Sciences**

STA 1202 – Introduction to Probability Theory  
Tutorial No: 1

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1. A bag contains 12 marbles comprising 3 black, 5 red and 4 green. One marble is drawn randomly from the bag.
  - a. Determine the probability of each of the colored marbles being drawn:
    - i. Black
    - ii. Red
    - iii. Green.
  - b. Show that the probabilities sum to 1.
  - c. What is the probability that the marble drawn is:
    - i. not Green?
    - ii. either Black or Red?
    - iii. Neither Red nor green?
    - iv. either Black, Red or Green?
2. A random experiment may result in 1, 2, 3, or 4, if  $Pr(1) = \frac{1}{13}$ ,  $Pr(2) = \frac{2}{13}$ ,  $Pr(3) = \frac{3}{13}$ , find the probability obtaining a 4.
3. A random experiment may result in A, B, C, D, E, where A, B, C, D, are equally likely and E is twice likely as A. find  $Pr(E)$
4. Fifty teenagers were asked what they did on the weekends. A total of 35 said they went to football matches, the movies or both. Of the 22 who went to football matches, 12 said they also went to the movies, show the information on Venn diagram. How many teenagers went to the movies but not football matches? How many did not go to either or these event
5.  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ ,  $A = \{1, 2, 3, 4\}$ ,  $B = \{2, 4, 6\}$  show the sets on the Venn diagram and your diagram to find:
  - a.  $(A \cup B)$
  - b.  $(A \cap B)'$
  - c.  $(A \cap B)$
  - d.  $(A \cap B)'$
6. Using the addition rule of probability,
  - a. If  $Pr(A) = 0.4$ ,  $Pr(B) = 0.7$  and  $Pr(A \cap B) = 0.2$ , find  $Pr(A \cup B)$ .
  - b. If  $Pr(A) = 0.6$ ,  $Pr(B) = 0.8$  and  $Pr(A \cup B) = 0.9$ , find  $Pr(A \cap B)$ .

7. X and Y are two mutually exclusive event of the random experiment.

$$Pr(X) = \frac{1}{6} \text{ and } Pr(Y) = \frac{7}{12}$$

Find the,

$$Pr(X \cap Y) \text{ and } Pr(X \cup Y)$$

8. X and Y are two independent events.

$$Pr(X) = \frac{1}{3} \text{ and } Pr(Y) = \frac{1}{4}$$

Find the,

$$Pr(X \cap Y) \text{ and } Pr(X \cup Y)$$

9. If  $Pr(A) = \frac{1}{2}$ ,  $Pr(B) = \frac{1}{3}$ ,  $Pr(A \cap B) = \frac{1}{6}$  find:

a.  $Pr(A \cup B)$

b.  $Pr(A | B)$

c.  $Pr(B | A)$

d. Check whether events A and B are mutually exclusive or not

e. Check whether events A and B are independent or not

10. If  $Pr(A) = 0.4$ ,  $Pr(B) = 0.5$ ,  $Pr(A \cap B) = 0.2$

a.  $Pr(X \cup Y)$

b.  $Pr(A | B)$

c.  $Pr(B | A)$

11. Three coins are tossed and the outcomes noted. List of the sample space of this experiment.

12. Nadia knows that if her car starts, she has an 80% chance of getting to work on time. However, if he car doesn't start, her chance of arriving on time is 50%. If Nadia's car starts only 70% of the time, what is the probability that:

- a. her car starts and she gets to work on time?
- b. she arrives at work late?
- c. she arrives at work on time?
- d. her car starts, given that she arrives at work on time?

13. Records indicates that, in Australia, 65% of secondary students participate in sport, and 71% of secondary students are Australian by birth. They also show that 53% of students are Australian by birth and participate in sport.

Referrer the table of Karnaugh maps and this information to find:

- a. the probability that a person selected at random is not Australian by birth
- b. the probability that a person selected at random is not Australian by birth, and does not participate in sport.

14. A drawer contains 7 T-shirts of which 3 are white and the rest are black. If 2 T-shirts are randomly selected from the drawer simultaneously, find the probability that they are:

- a. Both black
- b. Both white
- c. Different colors
- d. The same colors

**(Please answer question number 1, 2, 3, 4, 6, 7, 9, 12, 13, and 14 and submit the answer script to the tutorial room on or before 8<sup>th</sup> of November 2023)**